

Explainability and Causality Analysis

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Agenda

Introduction to Explainability

Types of Explainability

Feature Importance of inherently explainable models

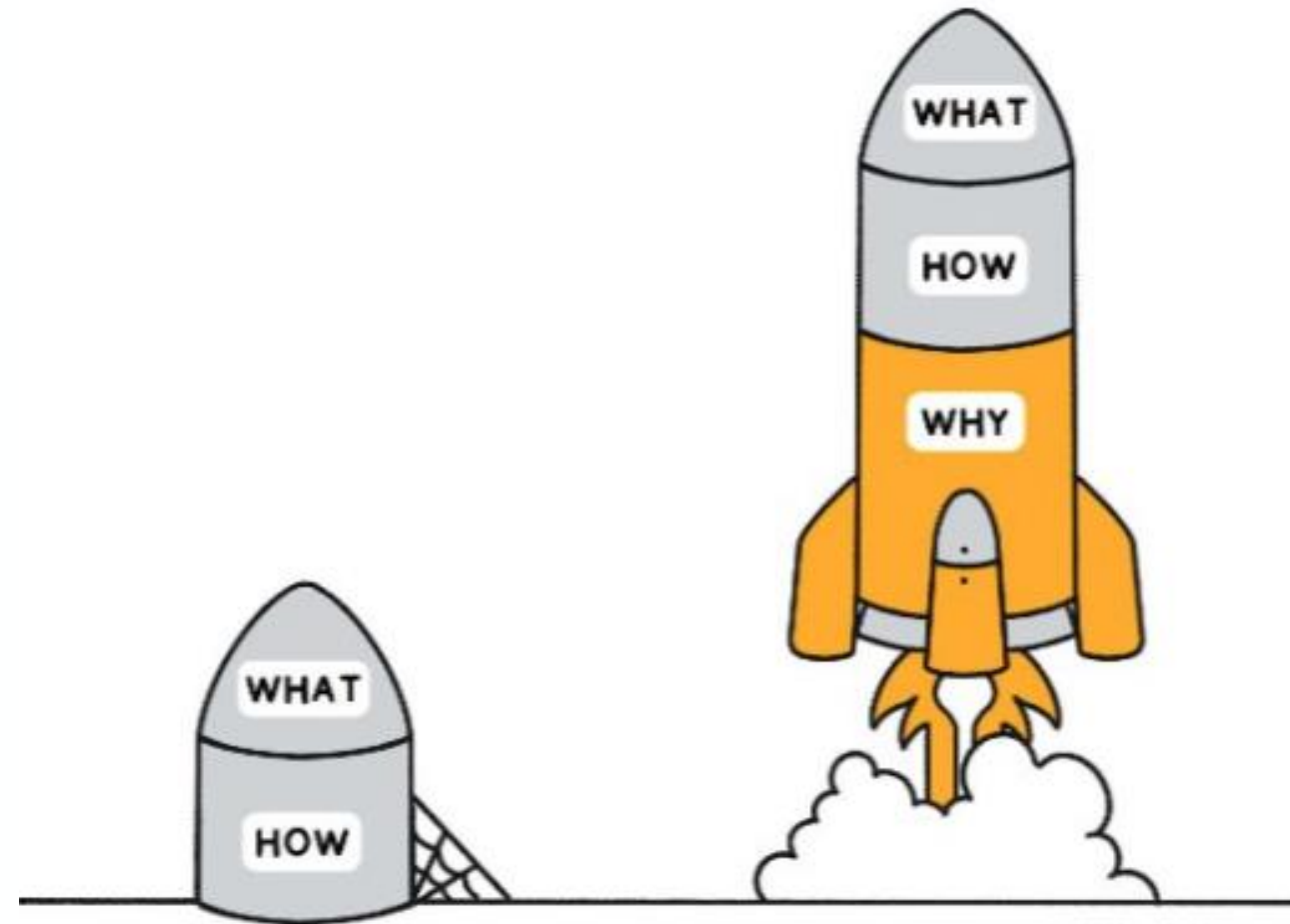
Permutation feature importance

Shapley values

Introduction to Causal Analysis



Explainability and Causality



Real-world Use-case

- **Scenario:** In a hospital, an AI model diagnoses a patient with vague symptoms and recommends an intensive, high-risk, and costly treatment.

Problem without Explainability:

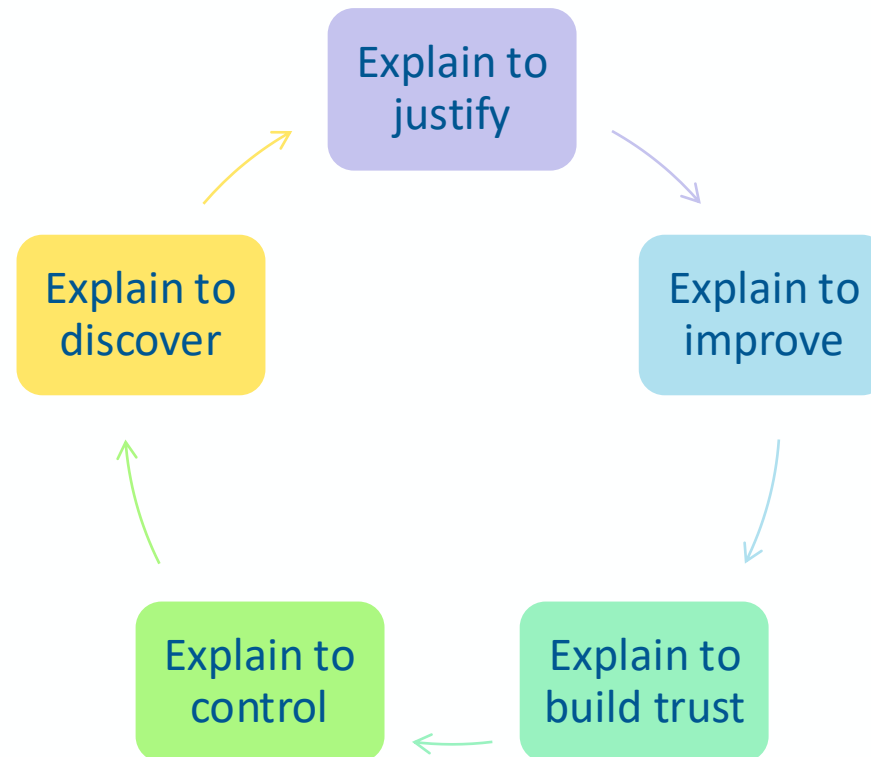
- The AI model gives no reasoning behind its recommendation.
- The doctor and patient are left in the dark—unable to understand the *why* behind the AI's suggestion.
- Trust is compromised, and the doctor can't fully justify or validate the treatment to the patient.

Explainable AI Solution:

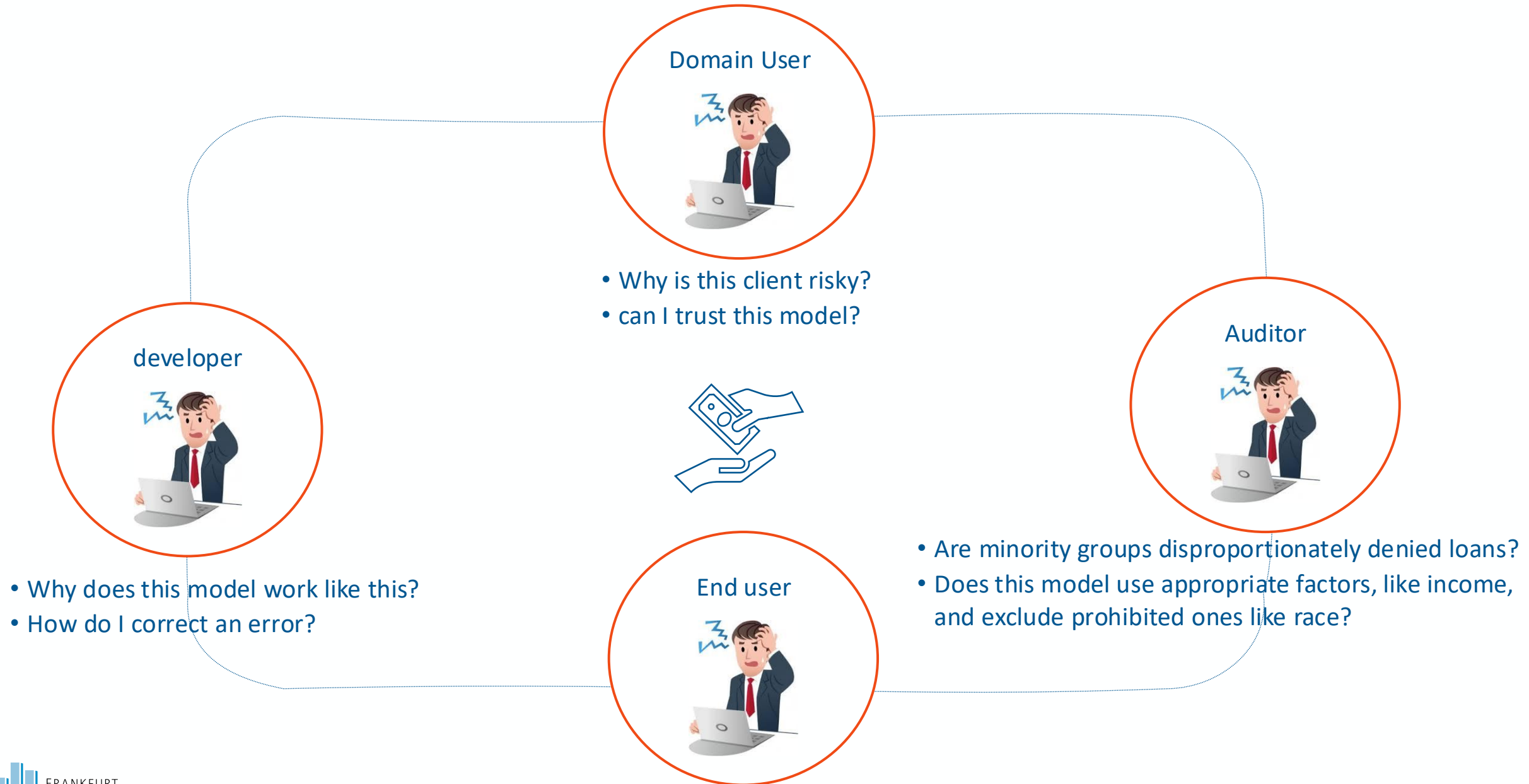
- The model highlights specific symptoms, risk factors, or patterns it used to arrive at its decision.
- Doctors can better understand the AI's logic, confirming if the recommendation aligns with medical judgment.
- Patients gain insight into their treatment, empowering them to make informed decisions and feel confident in their care.

Explainable modelling

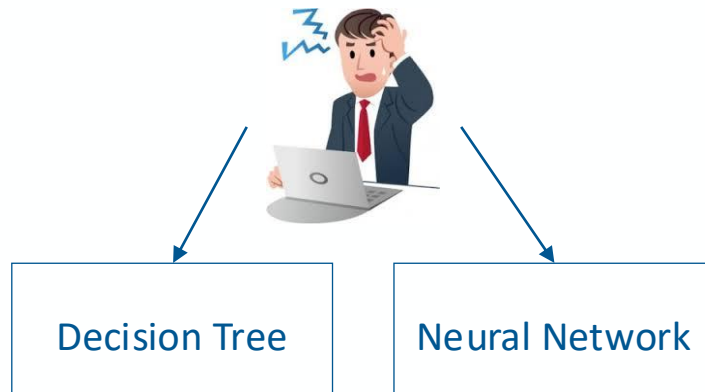
Explainability is about making model's decisions (especially AI-based models) **understandable to humans**.
It's not just *what* the model decided — it's *why* and *how*.



Different perspectives on why we need explainability: example for loan approval

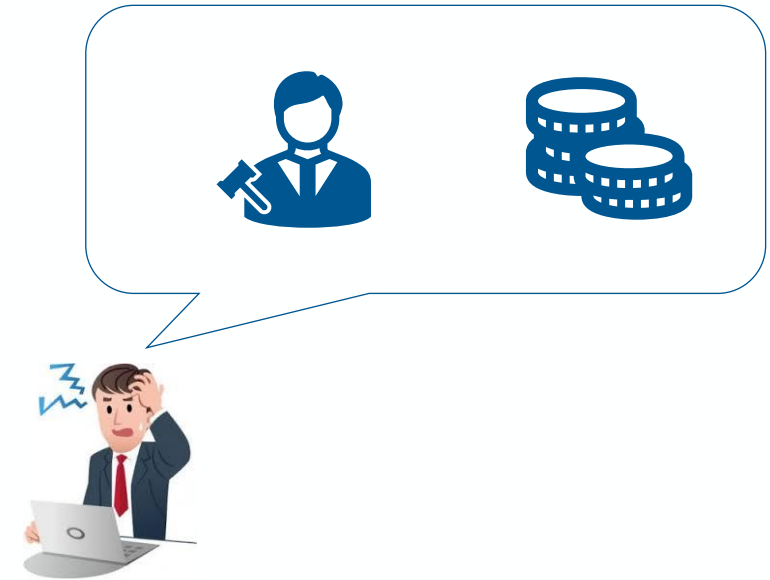


Trade-off between Explainability and Accuracy



Intepretibaility		
Accuracy		

Models that are highly accurate, such as deep neural networks, often lack transparency, while more interpretable models, like decision trees or linear regression, may sacrifice predictive performance for the sake of clarity and understanding.



Two key factors to consider when selecting a machine learning model are return on investment (ROI) and compliance. A company's ultimate goal is to maximize profit while adhering to legal and societal standards.

Explainability: Why It Matters More in Some Fields

Matters More When:

- Decisions affect **people's lives** (e.g., healthcare, finance, justice, Finance)
- **Regulations** require transparency
- **Trust and accountability** are critical

Matters Less When:

- Decisions have low risk (e.g., product recommendations)
- Speed or accuracy is more important than interpretability

Bottomline: Explainability is essential when the **stakes are high**—legally, ethically, or socially.



Transportation



Finance



Security



Legal



Medicine



Military

Fields where explainability is
crucial for trust and accountability

Feature importance

Often, the objective is to understanding What Drives a Target Variable and not to predict it.

- **Example use-cases:**

- Determine which of hundreds of financial indicators (e.g., transaction history, credit mix, payment patterns) are truly driving loan defaults.

- Identify key drivers from thousands of behavioral signals (e.g., clicks, scroll depth, session time) that predict conversion or churn.

- Understand which network traffic features indicate a potential cyberattack in high-dimensional system logs.

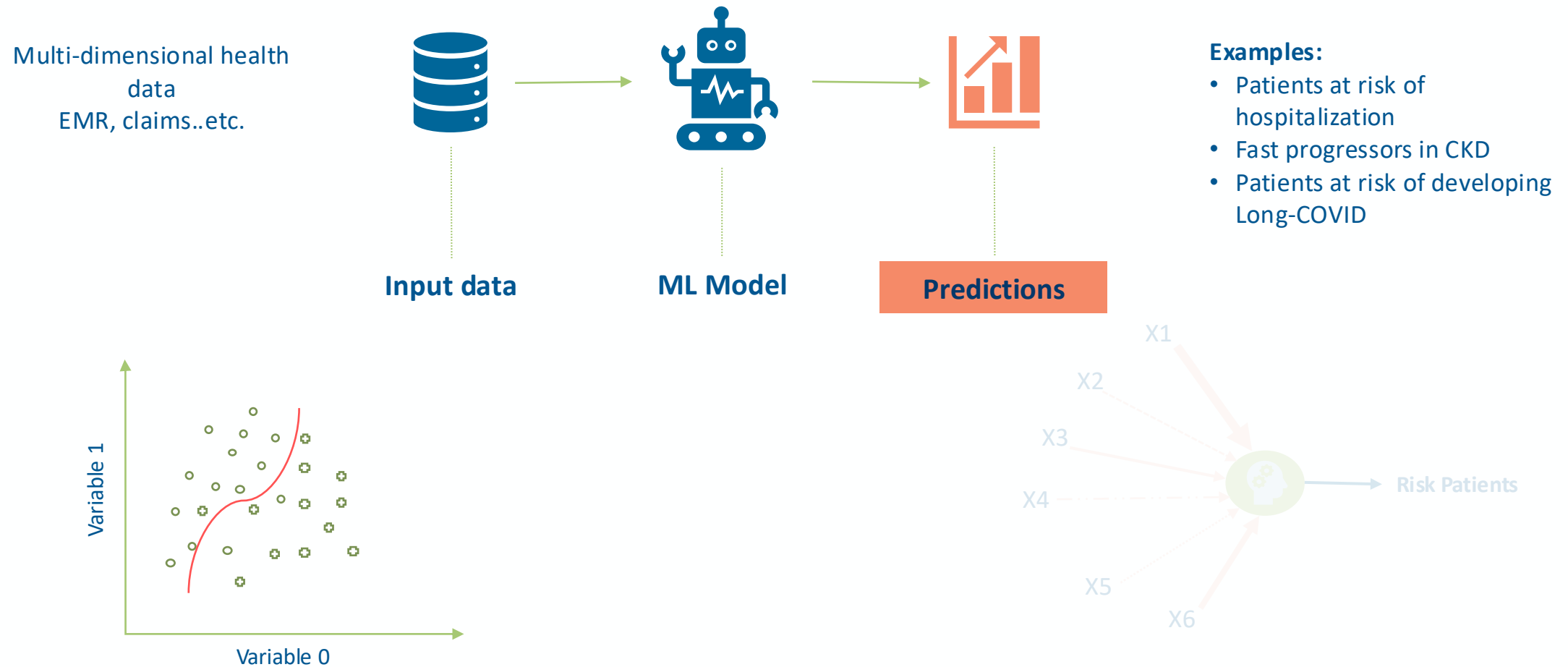
Feature importance

Often, the objective is to understanding What Drives a Target Variable and not to predict it.

Sometimes, we are more interested in **understanding the associations** between input features and a target variable than in simply predicting a class or running a regression model. **This is where Feature Importance Analysis comes in to highlights the most influential features in a model**

Use-case: Build a Model to predict diseases in order to understand the features that drive the risk.

Step 1: train a model

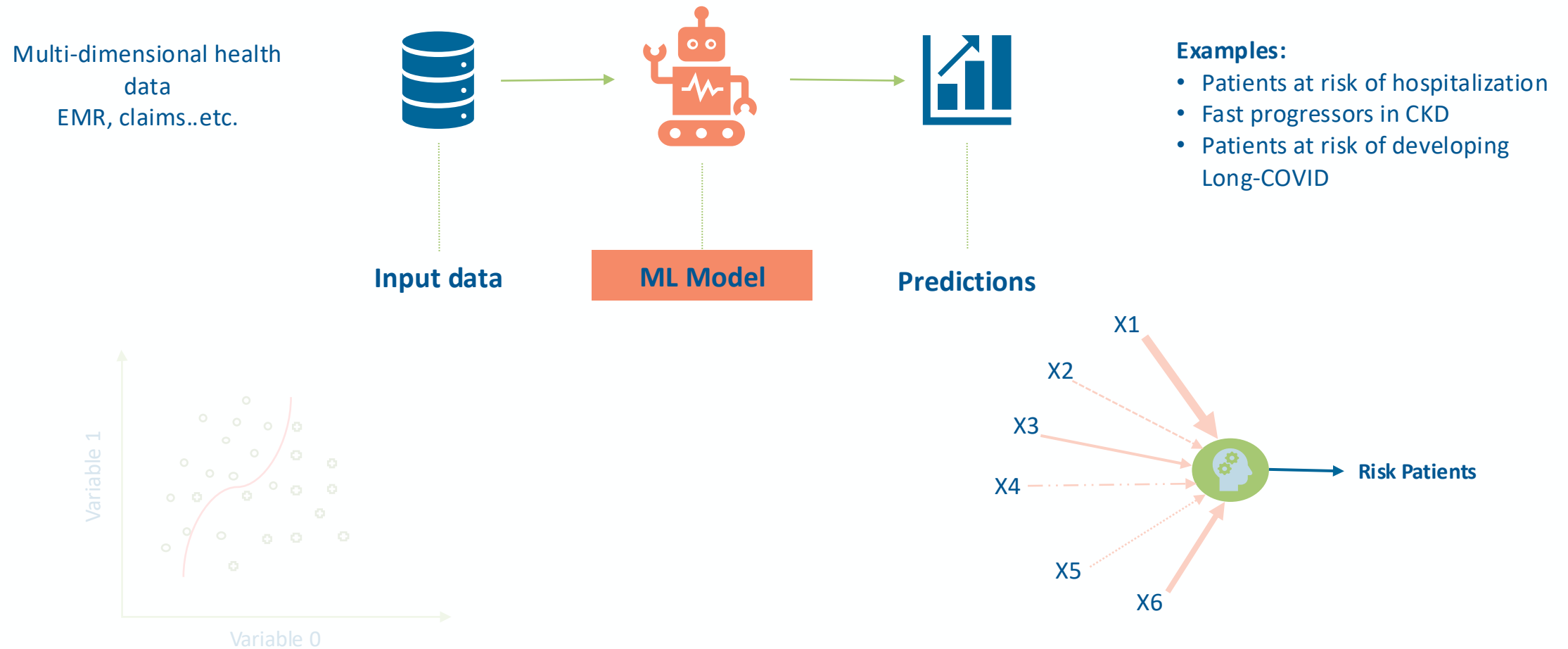


The model learns the risk factors and use them to predict patients with outcome



Use-case: Build a Model to predict diseases in order to understand the features that drive the risk.

Step 2: explain the model



Once the model is trained, we leverage it to get the risk factors the model used to predict patients with outcome

How can we explain the decisions of our AI?

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Global vs Local explainability

Local Explainability

Goal: Explain a **single prediction** made by the model.

Answers: *"Why did the model predict **this** for **this instance**?"*

Shows how each feature **contributed to a specific output**

Useful for:

- Case-by-case analysis (e.g., loan approval)
- Debugging unexpected predictions
- Communicating results to end users

Examples:

- SHAP Force Plot (for one prediction)
- LIME: Linear approximation around one point
- Counterfactual explanations: "What if income was higher?"

Global Explainability

Goal: Understand the model's **overall behavior** across all data.

Answers: *"What are the important features for the model?"*

Helps build trust and identify biases in the model

Useful for:

- Feature selection
- Model monitoring
- Documentation & auditing

Examples:

- Feature importance rankings
- SHAP Summary Plot
- Global Surrogate Models

Inherent vs. post-hoc Explainability

Inherent Explainability



Glass box models

Models are **transparent and self-explanatory**

You can directly inspect the logic or coefficients

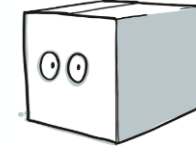
Examples:

- Linear Regression
- Logistic Regression
- Decision Trees (shallow)
- Rule-based models

These models are **inherently interpretable**—we don't need extra tools to understand their decisions.

May sacrifice predictive **accuracy** on complex tasks

Post-hoc Explainability



Black box models

Models are complex or opaque

You need separate techniques to **interpret** their behavior

Examples:

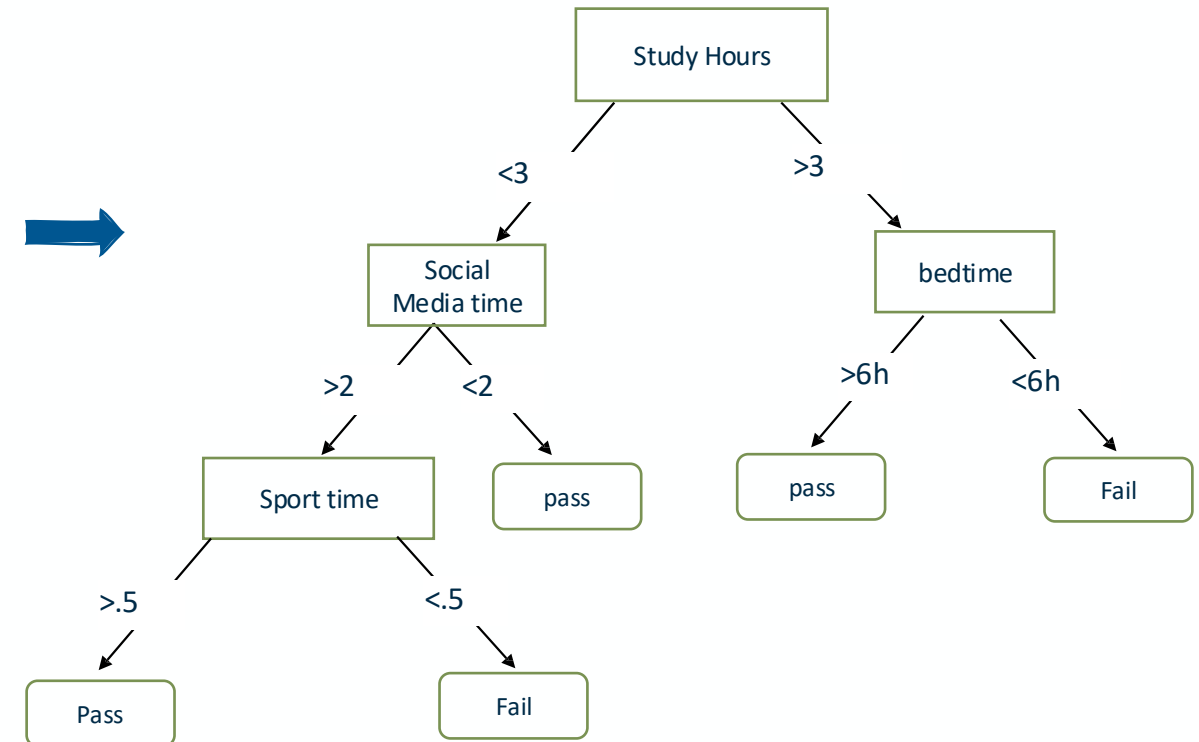
- Neural Networks
- Random Forests
- Gradient Boosting (e.g., XGBoost)
- Support Vector Machines in high dimensions

These models are not interpretable by default, so we use **Post-Hoc Explainability tools** like: SHAP, LIME...etc.

Maintains **performance** while offering insights

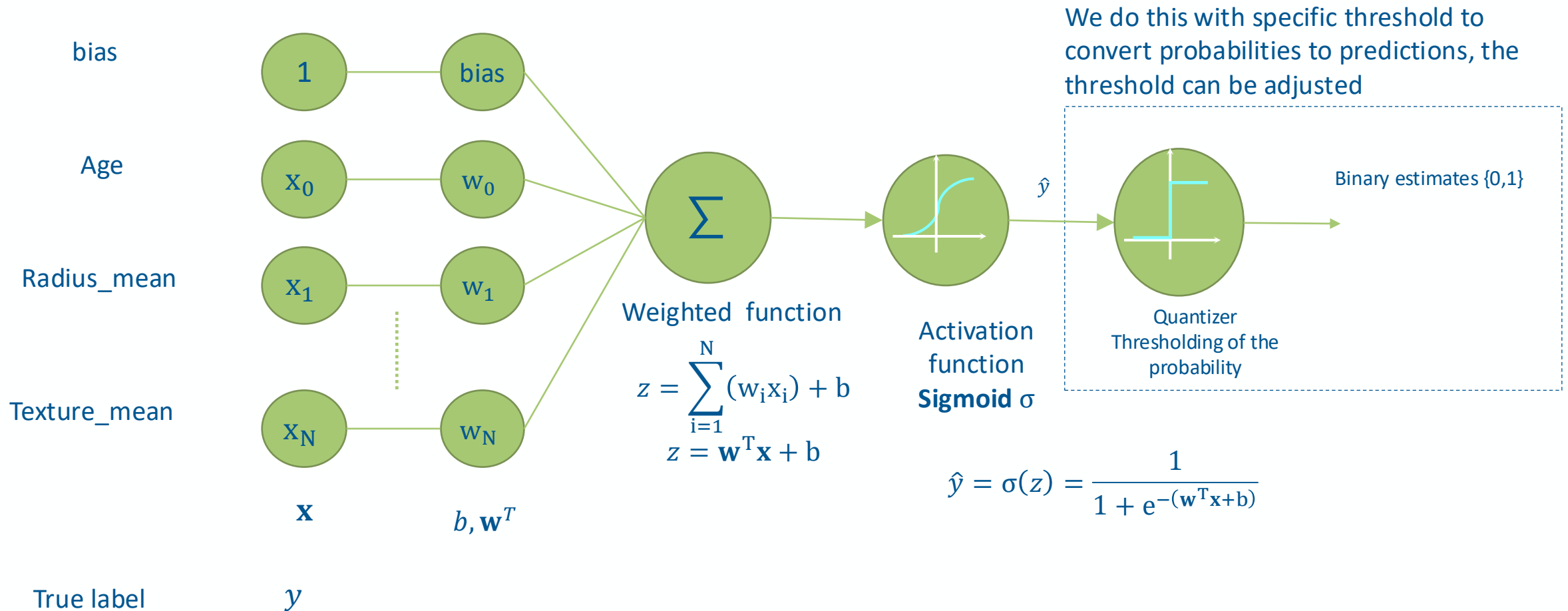
Inherently Explainable models: Decision Tree

Name	Study Hours	Social Media Time	Sport Time	Bedtime	Age	Result
...	...	...	...	...	...	...



One can use the order of the splits to infer and order most important features for the prediction

Inherently Explainable models: logistic regression



Logistic Regression to predict breast cancer using features extracted from pathological images

The learnt weights can be used to infer the logic the model used to predict cancer

Blackbox methods

While each tree is an explainable given the subset of the features and samples it uses, the decision made by the collection of these trees is difficult to explain. However, post-hoc methods could be used to explain

Mass (g)	Color	Texture	pH	Label
84	Green	Smooth	3.5	Apple
121	Orange	Rough	3.9	Orange
85	Red	Smooth	3.3	Apple
101	Orange	Smooth	3.7	Orange
111	Green	Rough	3.5	Apple
...				
117	Red	Rough	3.4	Orange



Bagging +
Random Subspace Method +
Decision Tree Learning Algorithm



Post-hoc Explainability: Feature importance via permutation

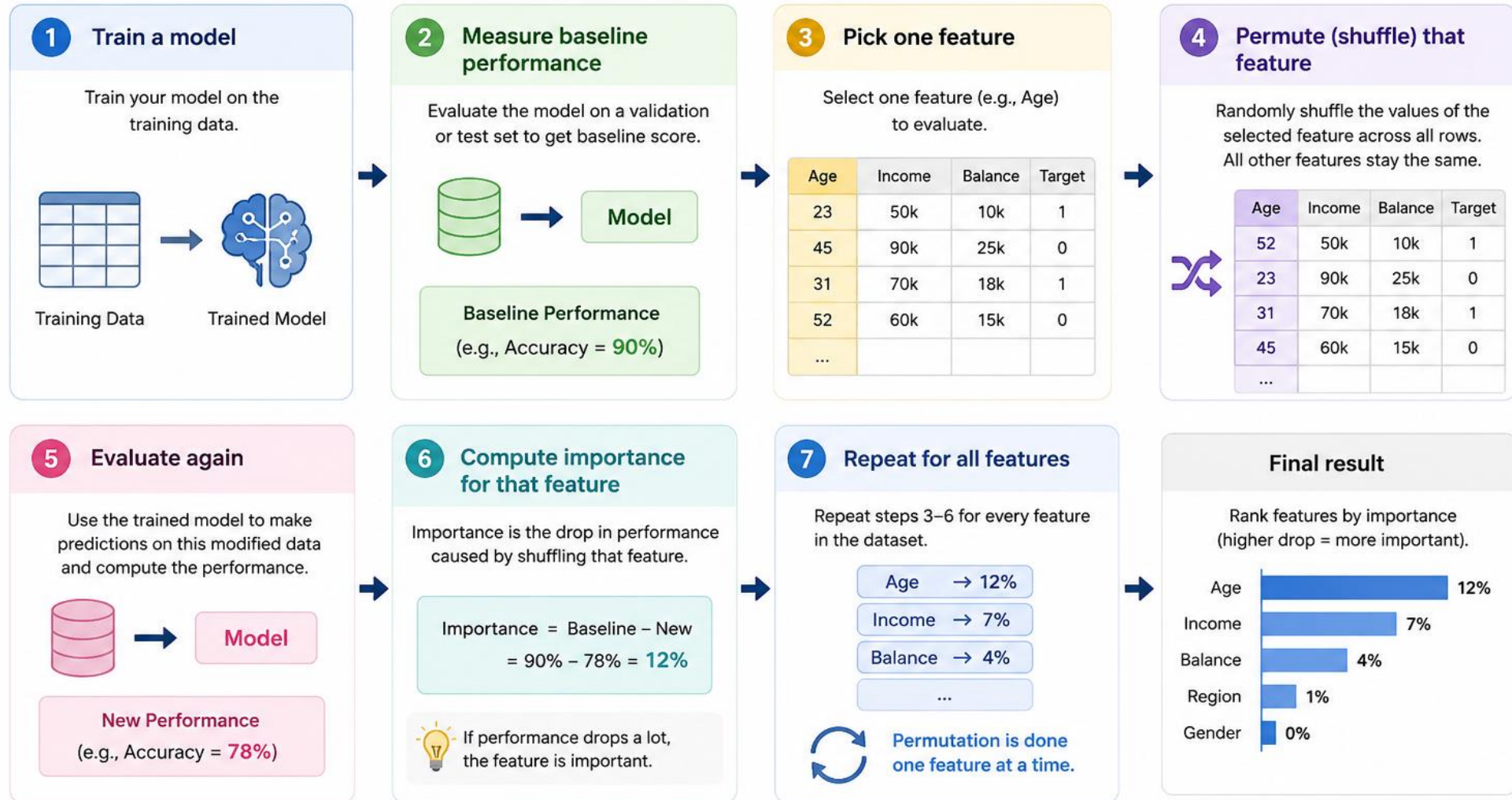
Permutation Feature Importance measures how much a model's performance drops when the values of a single feature are randomly shuffled.

Underlying hypothesis:

If a feature significantly contributes to the prediction, then permuting its values should result in a proportional decrease in model performance. Since permutation will **break the relationship** between the feature and the target.

PERMUTATION FEATURE IMPORTANCE: THE ALGORITHM

Measure how much model performance drops when each feature is randomly shuffled.



Key idea: If the model relies on a feature, breaking the relationship between that feature and the target (by shuffling it) will make the model perform worse.

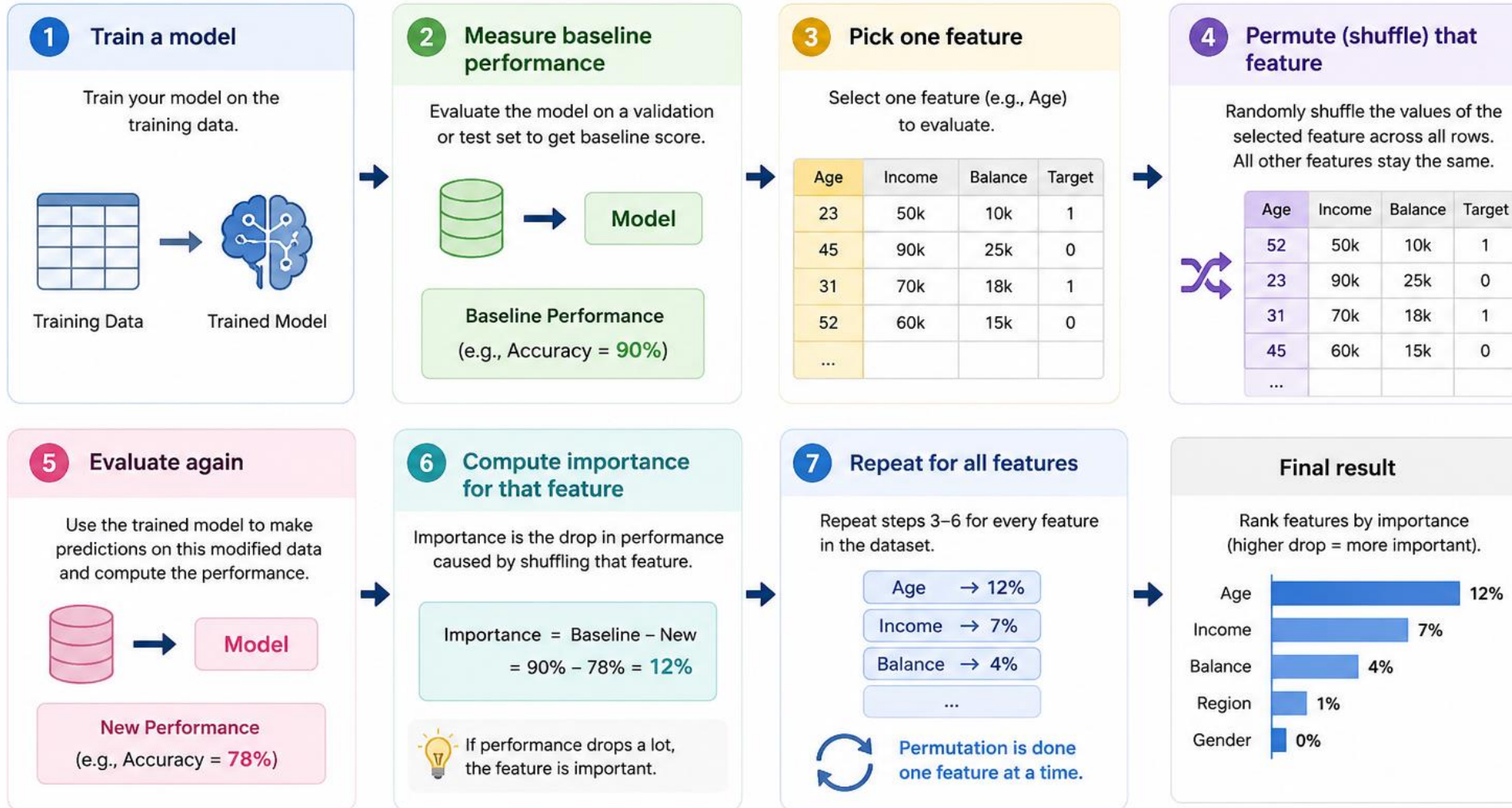
Quiz

After running the algorithm, I figured out that permutating a specific feature is slightly increasing performance, why may this happen?

Negative permutation importance does not mean the feature is useless. It means the model performs better when the feature's information is destroyed, suggesting noise, overfitting, redundancy, leakage, or a harmful learned dependency.

PERMUTATION FEATURE IMPORTANCE: THE ALGORITHM

Measure how much model performance drops when each feature is randomly shuffled.



Steps 4-6 are repeated K times to account for random variation

Summary of Feature importance via permutation

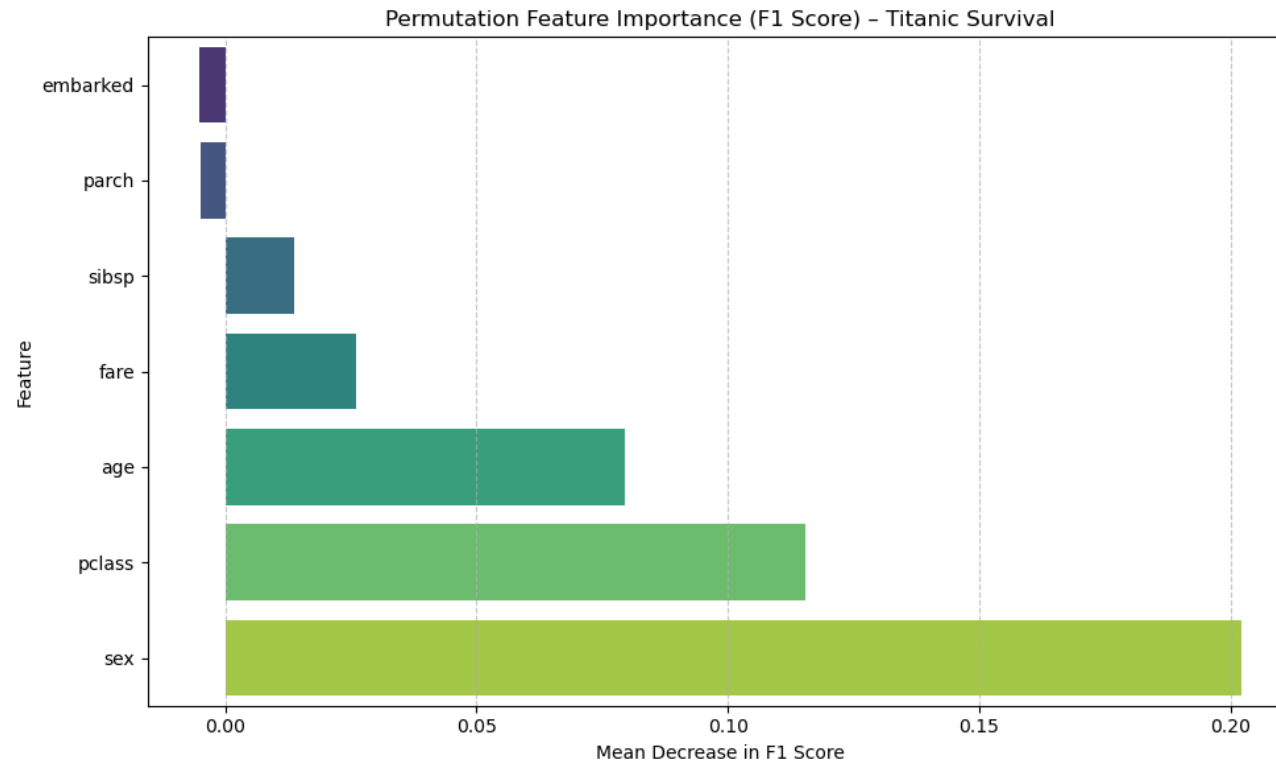
- **Train the model:** Fit your model (e.g., RandomForestClassifier) on your training data.
- **Evaluate baseline performance:** Use a **scoring function** (e.g., accuracy, R^2) on a **held-out set** to compute the performance score s before permutation.
- **For each feature:**
 - **Repeat K times:** the following is repeated multiple times to account for **random variation**, and the importance scores are averaged.
 - Randomly **permute (shuffle)** the values of that single feature **across all samples** in the validation set.
 - Compute the model's performance on the **modified dataset**.
 - Calculate the **difference** (or drop) in performance $s_{k,j}$ compared to the baseline.
 - Calculate the **difference between the average** drop in performance across all repetitions compared to the baseline.

$$i_j = s - \sum_{k=1}^K s_{k,j}$$

Post-hoc Explainability: Feature importance via permutation

```
# Train a model
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)

# Compute permutation importance
result = permutation_importance(model, X_test, y_test, scoring='f1', n_repeats=10, random_state=42)
```



Single-permutation-based variable-importance measures for the explanatory variables included in the random forest model for the Titanic data using F1 score as the scoring function.

Pros, Cons & Use Cases of feature permutations

✓ Advantages:

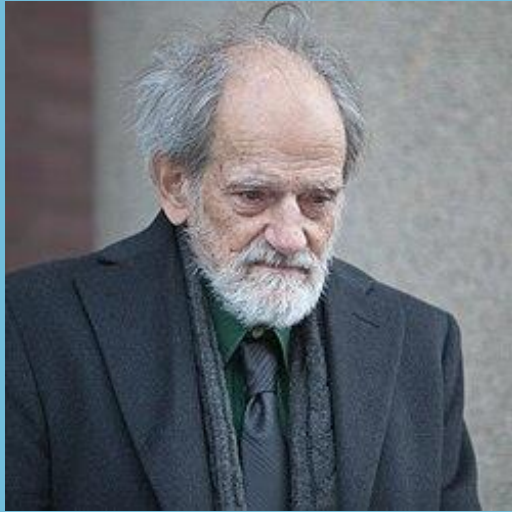
- Model-agnostic
- Easy to explain and implement

✗ Limitations:

- **Correlated features** can mask each other's importance
- Computationally expensive (especially with many features)
- Can be **unstable** on small datasets or high-variance models



Post-hoc Explainability: Feature importance via SHAPLEY Values



Lloyd Shapley in 2012, [Nobel Memorial Prize in Economic Sciences](#)

- Originally used to fairly distribute **payouts (total gains)** among a group of players who have collaborated in a **cooperative game**.
- In ML, "players" = **features**, "payout" = **model prediction**
- **Goal:** Quantify **each feature's contribution** to a specific prediction.
- In ML is it called SHAP Value (SHapley Additive Explanations)

Post-hoc Explainability: Feature importance via SHAPLEY Values

Method Overview

- Imagine you're trying to **divide the model's prediction** fairly among the input features:
 - You consider **every possible subset** of features **excluding** feature i .
 - For each subset S , compute the marginal contribution of feature i to the subset (**how much feature i improves the prediction** when feature i is added to S):

$$f(S \cup \{i\}) - f(S)$$

- You **average** this marginal contribution over **all possible combinations**, weighted to reflect how many times each subset appears in permutations.
- This ensures:
 - Features that **consistently add value** get a high Shapley score.
 - Features that only sometimes help get proportionally less credit.

SHAP Values: Fair Feature Attribution

Example: What would be the importance of feature "fare" in predicting survival for the titanic Dataset.

Subset of the features S	$v(S)$	$v(S \cup \{\text{"fare"}\})$	Marginal Contribution $v(S \cup \{\text{"fare"}\}) - v(S)$
$\emptyset$	0.5	0.7	0.2
{age}	0.4	0.6	0.2
{sex}	0.6	0.8	0.2
...	...	...	...
{age,sex, sibsp}	0.65	0.75	0.10

$$\varphi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S)) = 0.1667$$

A combinatorial term that gives **fairness** by averaging over **all possible orderings** of features

Explaining the weighting schema of SHAP

Suppose the model predicts house price using:

Age, Income, Gender

Imagine building the prediction one feature at a time, who should get credit for the final prediction?

The problem is that a feature's contribution depends on **what information is already available**.

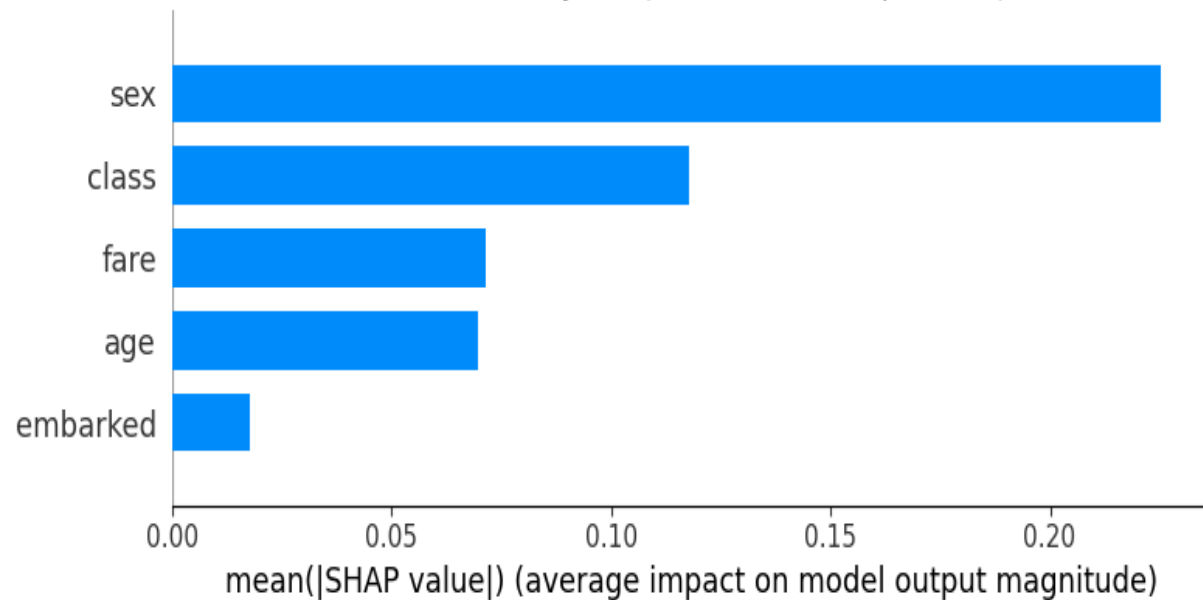
Understanding the weighting schema

Permutation	Features before Gender (S)	Marginal contribution of Gender
Gender → Age → Income	{}	$f(\{\text{Gender}\}) - f(\{\})$
Gender → Income → Age	{}	$f(\{\text{Gender}\}) - f(\{\})$
Age → Gender → Income	{Age}	$f(\{\text{Age}, \text{Gender}\}) - f(\{\text{Age}\})$
Income → Gender → Age	{Income}	$f(\{\text{Income}, \text{Gender}\}) - f(\{\text{Income}\})$
Age → Income → Gender	{Age, Income}	$f(\{\text{Age}, \text{Income}, \text{Gender}\}) - f(\{\text{Age}, \text{Income}\})$
Income → Age → Gender	{Income, Age}	$f(\{\text{Age}, \text{Income}, \text{Gender}\}) - f(\{\text{Age}, \text{Income}\})$

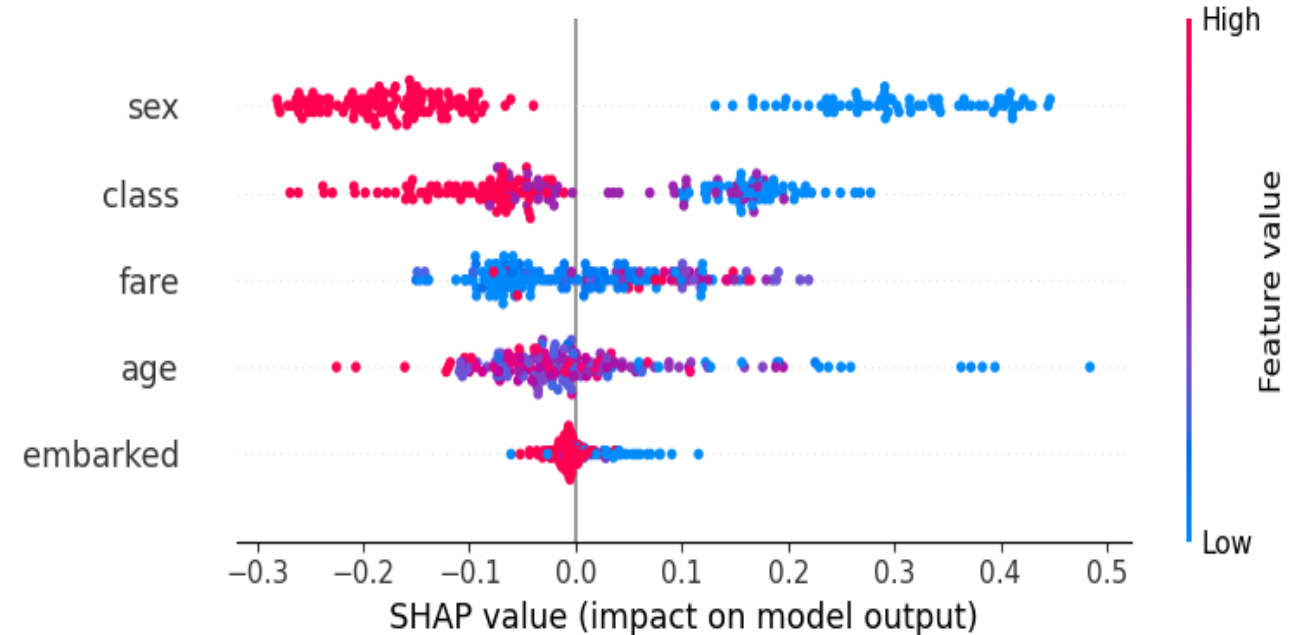
Global vs. local impact using SHAP python package

SHAP attributes weights to each feature regarding its contribution towards the prediction of an individual observation

SHAP Summary Plot (Global Feature Importance)



Average global impact on model output sorted per importance



“How much does each feature, for each data point, contribute to the model’s prediction compared to a baseline?”

Quiz

A highly accurate model predicts Titanic survival. Its explainability report (e.g., SHAP) shows that **Fare** is one of the most important features.

- a. Fare directly affects survival because changing an important feature should change the prediction.
- b. Fare is useful for prediction, but it may simply capture information about other factors that actually influenced survival.
- c. Fare must have been included in the evacuation policy, otherwise the model would not rank it so highly.
- d. Since Fare is highly important across many passengers, it is both predictive and causal.

Quiz

Imagine every passenger paid exactly the same fare, but everything remained unchanged. **What would you expect?**

- a. The survival rate would increase because Fare was an important feature.
- b. The model would become less accurate, but the actual survival outcomes would likely remain unchanged.
- c. The survival rate would become nearly identical across all passengers.
- d. It is impossible to say because feature importance measures causal effect.

- **Predictive importance** ("The model uses this.")
- **Actionability to know the causality** ("Changing this changes the outcome.")

Be careful with causal interpretations: SHAP values do NOT reflect CAUSALITY

Predictive models don't usually deliver causal insights explicitly unless modified and trained to do so

Interesting Reading on this topic: [Be careful when interpreting predictive models in search of causal insights](#)

Causality Analysis



- **Causal Analysis** focuses on understanding **cause-and-effect relationships** rather than just correlations. It goes beyond predictive models by identifying what *causes* certain outcomes, which is crucial for decision-making, policy setting, and understanding interventions.
- **Key Concepts:**
 - **Causal Inference:** Determining if one event or variable causes another.
 - **Counterfactual Analysis:** Asking "what if" questions to assess outcomes under different scenarios.
 - **Interventions:** Simulating or applying actions to see their causal impact.
- **Example Use Cases:**
 - **Healthcare:** Identifying the effect of a treatment on patient outcomes.
 - **Economics:** Analyzing the impact of policy changes on economic indicators.

Background and related questions

Decision-making requires understanding interventions, not just model reasoning

What is the likelihood this patient with BREAST CANCER TREATMENT, will survive 5 years?



- What if we must decide to start an aggressive treatment based on the predicted outcome?
- What if the patient has developed unrelated comorbidities in-between and that those led to death?
 - What if the patient's lifestyle becomes unhealthy after the treatment?

Background and related questions

Decision-making requires understanding interventions, not just model reasoning

What is the likelihood this patient with BREAST CANCER TREATMENT, will survive 5 years?



Other questions:

- Does smoking cause breast cancer?
- Does obesity cause low response for drug A?
- $\Pr(\text{lung cancer} | \text{smoker})$ vs $\Pr(\text{lung cancer} | \text{nonsmoker})$

Why cannot we simply use predictive modeling or simply correlation for this?

- Explainability reveals **what influenced the model**, not **what changes the outcome**.
- Explainability of predictive models only are like correlations. They may also attribute importance to **confounders**—hidden variables that create misleading correlations.
- Causality accounts for confounders to separate true causes from misleading correlations.

Examples

1. Model Explanation:

- Customers who own umbrellas have a higher likelihood of traffic accidents
- The model identifies *umbrella ownership* as an important feature

2. Causal Reality:

- **Confounder:** Rain
- Rain increases both umbrella usage and accident risk
- Giving everyone an umbrella will **not** reduce accidents

Example: Gender (Sex) vs. Survival of the Titanic sinking

1. **Explainability:** The model says **Sex** is the most important feature because women survived much more often.
2. **Causality:** Being female itself did not biologically cause survival. The underlying cause was the "**women and children first**" **evacuation policy**, which determined access to lifeboats.
3. **Confounder:** Evacuation policy (not directly recorded in the dataset).

Example: Fare vs. Survival of the Titanic sinking

1. **Explainability:** Higher **Fare** is an important predictor of survival.
2. **Causality:** Paying a higher fare did not directly increase survival. Higher fares were associated with **higher passenger class**, which meant cabins were closer to lifeboats and passengers had priority access.
3. **Confounders:** Passenger class, cabin location, crew assistance.

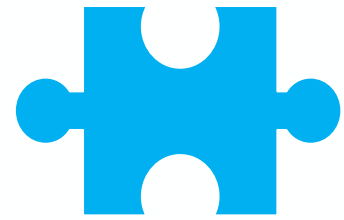
Summary: Explainable vs. Causal AI

Explainability Analysis

- Focuses on making models' decisions and processes more understandable to humans.
- Provides reasoning behind predictions based on patterns in the data, often using methods like feature importance, attention maps, or decision trees.
- Answers the question: *"What factors influenced this particular decision?"*

Causal Analysis

- Aims to uncover cause-and-effect relationships, going beyond correlation to determine *why* changes in one variable led to changes in another.
- Uses causal inference techniques, often relying on domain knowledge, experiments, or observational data to validate real-world causal links.
- Answers the question: *"What would happen if we intervened or changed a specific factor?"*



Questions



Answers

